

Multi-Phase Results and Cross-Phase Analysis

Phase-Specific Findings I

Foundation Phase (Phase 1)

The configured foundation phase retained `{{PHASE_1_PAPERS}}` records after its search, filtering, and de-duplication rules. Its taxonomy includes transit spectroscopy, emission spectroscopy, phase-curve analysis, and direct imaging. These labels describe the configured retrieval design; the synthetic fixture cannot establish how widely those approaches occur in the published literature.

JWST Phase (Phase 2)

Phase 2 retained `{{PHASE_2_PAPERS}}` records under the JWST-oriented queries and the configured year filter (2020). The boundary is intended to include pre-launch calibration work and post-launch observations. It is a design choice, not evidence that the phase is exhaustive or that any instrument caused a change in the field.

Phase-Specific Findings II

Molecular Detection Phase (Phase 3)

Phase 3 retained `{{PHASE_3_PAPERS}}` records under five configured molecule-oriented query groups: H O, CO , CH , H S, and Na/K. The query groups operationalize the review question; they do not assert that these are the most commonly detected species or that the fixture establishes a scientific trend.

Knowledge Graph Results I

The optional knowledge-graph stage reports

`{{TOTAL_ASSERTIONS}}` assertions from the eligible sample.

The configuration-derived hypothesis table is the authoritative mapping from review question to score:

`{{HYPOTHESIS_TABLE}}`

The configured scores are descriptive evidence summaries, not calibrated probabilities, causal effects, or conclusions about instrument performance. The table should be read together with the claim ledger and source-tier metadata. A fixture run cannot establish whether models and observations agree in the broader literature.

Citation Network Analysis I

The combined corpus contains $\{\{\mathbf{CITATION_EDGES}\}\}$ citation relationships across $\{\{\mathbf{CITATION_NODES}\}\}$ papers, with a network density of $\{\{\mathbf{CITATION_DENSITY_PCT}\}\}\%$. This is a descriptive statistic for the retained corpus; it does not establish a field-wide citation structure or community claim.

Reproducibility Assessment I

Reproducibility assessment of the sampled papers revealed a mean composite reproducibility score of $\{\{\text{REPRO_MEAN_SCORE}\}\}$, with $\{\{\text{REPRO_PAPERS_SCORED}\}\}$ papers successfully scored. This analysis examines the computational workflow transparency of each paper, evaluating source data availability, method documentation, experimental reproducibility, and output accessibility.